HW2: Digital PLL homework assignment

Due Date: 16 March

I. Introduction

This assignment brings together what you have learned in the class so far. It explores both s-domain and time-domain modeling of an ADPLL to obtain and validate the performance of the frequency synthesizer for wireless applications.

For the deliverables, submit them on Brightspace in a zip file containing the report, the MATLAB code (.m file), and any other relevant material. Please, name your zip file as XXXXXX_Surname_Name_Assignment#2.zip, where XXXXXX is the student number.

II. Assignments

Part 1: S-domain simulation

a) Investigate the s-domain ADPLL loop, taking into account a crystal REF of 100 MHz with a phase noise of -155 dBc/Hz, over the entire spectrum, and a free-running DCO phase noise of -120 dBc/Hz at 1 MHz offset (in 20dB/dec region), for a central frequency of 2.4 GHz. This is typical for an LC Oscillator, without taking into consideration the upconverted flicker noise. For the in-band noise, consider the TDC resolution of 2ps.

b) Based on the values given in item a), find the loop filter coefficients (α, and ρ) in order to **minimize** the integrated jitter. Explain the reasons for your choices.

c) What is the phase margin and gain margin of the given ADPLL? What is the closed-loop bandwidth?

d) Does the type-II operation help in any sense for the integrated jitter? Is the loop stable in these conditions?

e) Plot the transfer function of the closed-loop system using all the given parameters and the coefficients found at previous items. Verify the choices you made in item b) and draw your conclusions.

f) Calculate the following parameters:

- DCO natural phase noise contribution
- TDC phase noise contribution
- Reference phase noise contribution
- Phase noise contribution of DCO frequency quantization

Part 2: Reference and oscillator time-domain simulation

a) Develop a time-domain model for the reference oscillator with the center frequency and phase noise mentioned in part 1, and plot a phase noise plot.
b) Develop a time-domain model for the RF oscillator with the center frequency and phase noise mentioned in part 1, and plot a phase noise plot.

Part 3: ADPLL time-domain simulation

a) In order to validate the s-domain analysis, make a time-domain event-driven model ADPLL [1], similar to that depicted in Fig. 1. Consider K_{DCO} = *25kHz/bit.*

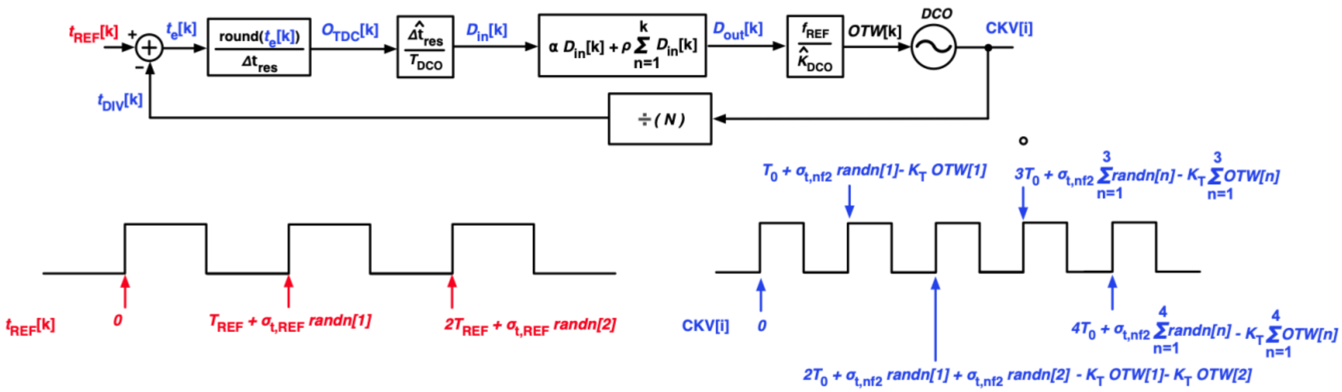

Fig.1 Digital PLL

b) Simulate the ADPLL model and plot the transient behavior of the loop, considering the initial oscillation frequency 20 MHz off the desired frequency and the parameters found in Part 1. Do the following:

- Plot 1: Instantaneous frequency deviation of CKV and identify the settling time.
- Plot 2: Instantaneous phase error, $\Phi_e[k]/2\pi$, obtained from the phase detector.
- Plot 3: Instantaneous period deviation of the CKV.

c) Plot the phase noise and explain the result. How does it compare to the transfer function found at Part A, item e)?

d) Calculate the integrated jitter when the system is already in lock.

Reference:

R. B. Staszewski, C. Fernando and P. T. Balsara, "Event-driven Simulation and modeling of phase noise of an RF oscillator," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 52, no. 4, pp. 723-733, April 2005, doi: 10.1109/TCSI.2005.844236.